Real-Time Embedded Systems

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Computer system

- Machine: Physical system to perform certain function
  - energy to operate
  - certain input
  - expected output
- System: a group of interacting/interrelated elements which follow rules to perform a certain dedicated task
- Computation: Mathematical operations: +, -, / ...
- Computer system is a machine which performs computation

Turing Machine

- The most fundamental definition of computer
- Tape: Divided into cells, each cell contains a symbol
- Head: Can read and write symbols
- State Register: stores the current state of machine
- Table: of instructions
  - Erase/Write symbol
  - Move head
  - Assume same or new state

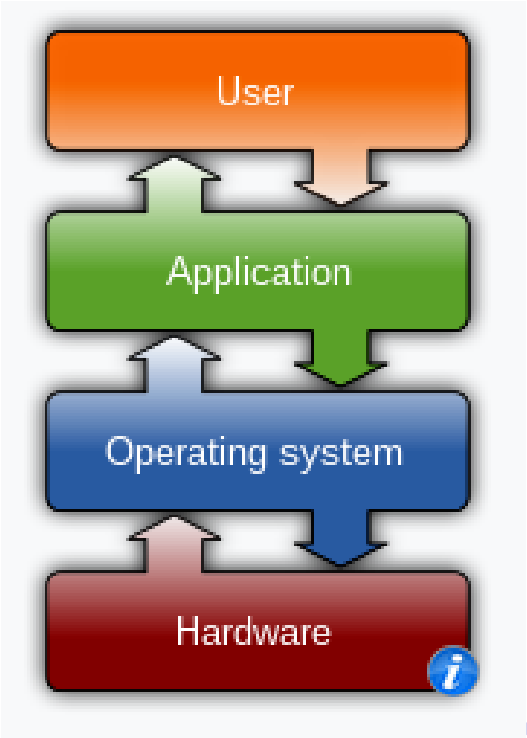
Modern Computer Systems

- Have become a lot more complex
- The usage has also become complex
- Multi-core, large memory areas, with small physical size
- Given way to System On Chips (SoC)
- Microcontrollers have become smaller
- It is generally difficult to impart the need to understand real-time aspects of embedded systems, especially if there is novelty

Operating System

- Software that manages computer hardware and software
- Provides services computer programs: game needs wifi, meeting eed camera, etc.

- Single Tasking OS: can only do one task at a time
- Muti-tasking: Tasks are scheduled
- OS is different from Basic Input Output Service (BIOS)



REAL TIME EMBEDDED SYSTEMS

Definition

- Embedded System: A computer system which is designed and deployed for a dedicated functionality.
- Real-Time Embedded System: an embedded system which needs to function within certain real world timing constraints or deadlines.
- Real world deadline:
- Clock: measures passage of time
- Time: 1 sec = caesium 133 atom 9192631770 radiation waves
- Eg: Coffee machine Dedicated task: to make coffee, should make coffee in 1 min, not 2 days ! (in human terms)

Hard, Soft Real-Time Systems

- Hard: RT system which if does not produce desired functionality before the deadline at any point in its life, leads to catastrophe.
- Eg.: Aircraft Control system, actuation should happen when there is pilot input
- Soft: R/T system which if does not produce desired functionality before the deadline, does not lead to catastrophe but a quality degradation, data after deadline can have some use
- Eg.: Video streaming, losing some frames leads to bad quality
- Firm: Results after a deadline does not harm but it is of no use to system

Murphy's Law

- "Anything that can go wrong will go wrong"
- Mrs Murphy's Law: "Anything that can go wrong will go wrong, while Mr Murphy is out of town"
- Apply to RT system: If your embedded RT system's functioning can go wrong, it will !!!
- QUESTION: HOW TO MAKE SURE IT DOES NOT ?

A man drowned crossing a stream with an average depth of 6 inches

Real-Time System Model

- Workload are modelled as task τ_i , can be a program code or a function
- Tasks executes on resource(s), processor(s)
- Task have an execution time c_i
- $c_i = f$ (hardware, logic, code, compiler...)
- Tasks have a Worst Case Execution Time (WCET) C_i
- Tasks have a timed deadline D_i
- Tasks can have priorities p_i
- Tasks can have period of execution T_i, at which they execute again
- A real Time System S is tuple

$$S = \{\tau_i, C_i, D_i, T_i, p_i\}, i = 1, 2, 3...n$$

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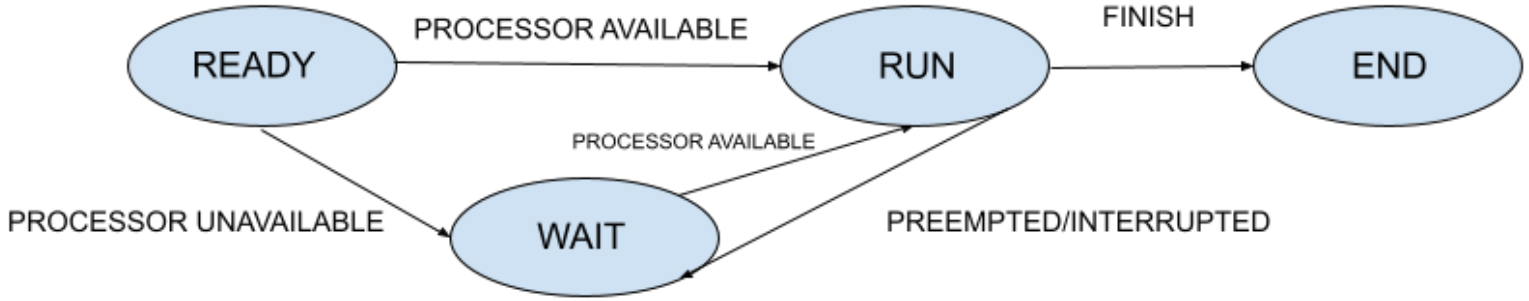
Task Execution Model

A task is in one of the following states

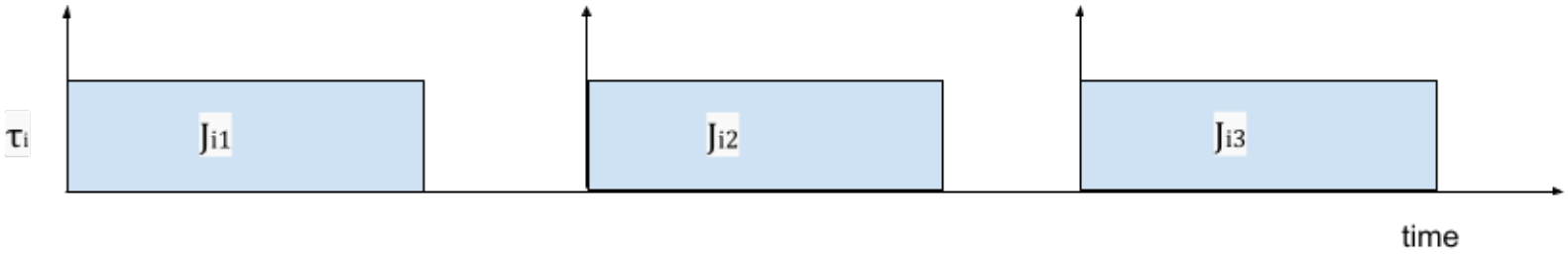
- A task is released and is *READY* for execution
- Task executes in *RUN*
- Task can *WAIT* for execution, it can happen immediately after release or it can be interrupted while execution
- Task *END* execution

The relationship between these states is:



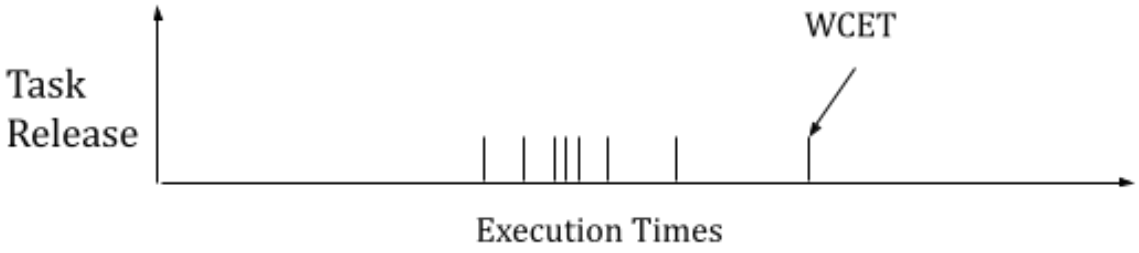
Job

- If task is periodic, each periodic instance of the task is a Job
- for i-th Task τ_i, j-th job is J_{ij}



Worst Case Execution Time

- WCET: the maximum amount of time a task can take to execute on a particular hardware
- Task takes certain time to finish execution
- This execution time is not deterministic, it can vary
- Variation can come from many sources: environment, other tasks, precedence constraints, memory access problems
- Task will never exceed WCET, it is the most pessimistic estimate



Priorities

- Tasks can have relative importance over others
- For our discussion, $p_i = 0$ is the highest
- Task with higher priority can arrive later than one lower priority already executing
- Scheduling decisions can use this priority

Resources

- Processors, memories, buses are resources
- Resources are used to execute tasks successfully
- Processors: executes instructions; single core, multi core
- Memories: allow data storage and access; cache, ram, rom, etc.
- Buses: allow data transfer
- All of them influence WCET of a task

Periodic, Aperiodic or Sporadic Tasks

- Periodic tasks are those which need to execute at certain intervals
- Eg.: Temperature sensor and calculations
- Aperiodic tasks are those which only execute once when called
- Eg.: coffee machine, tasks involved execute once to make a cup of coffee
- Sporadic tasks are this which can arrive any point in time
- Eg.: Sudden proximity alert in car while driving (Sporadic tasks are highly undesirable System has to keep room for them all the time, while they arrive only a few times, resources are wasted)

Real Time System Scheduling

- Tasks need time on resources to produce results
- Scheduling: Scheme of allowing time on resources to tasks for their execution so that every tasks is able to execute without any missing its corresponding deadline
- Time on resources corresponds to real world clock
- Scheduling is the act of governing tasks executions

Scheduling Types

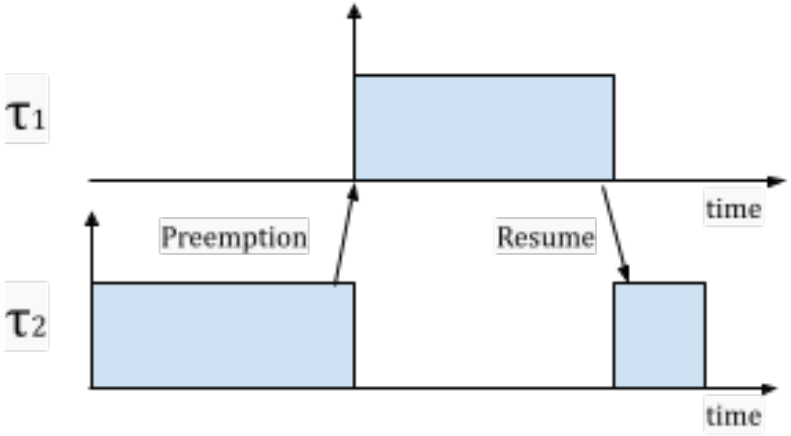
- Static vs Dynamic: Scheduling decisions are predetermined in static, dynaminc schedulers resource allocations can adjust depending on real-time demands
- Offline vs Online: Offline schedulers have predetermined rules, online can change those rules at runtime
- Preemptive vs Non-preemptive: Preemptive schedulers allow task interruption if another high priority task arrives, interrupted task goes to *Wait*
- Optimal vs Heuristic: Optimal algorithm minimizes some cost function; heuristic follows a guiding function which lay not be optimal

Schedule

- The policy/algorithm of allowing task execution
- Most used ones:
  - Fixed Priority (FP): Task priorities are fixed, highest priority task executes
  - Round Robin (RR): Rondo scheme, each task takes turn to execute
  - Rate Monotonic (RM): Task with high frequency (shortest period) has highest priority
  - Earliest Deadline First (EDF): As task jobs arrive, one with earliest deadline is executed first

Scheduling decisions: Preemption

- Each of these scheduling policies can be preemptive or non-preemptive
- Preemption is a scheduling choice, not task choice
- Preemption example:



Schedulability

Given a real Time System S is tuple

$$S = \{\tau_i, C_i, D_i, T_i, p_i\}, i = 1, 2, 3...n$$

- is it schedulable
- is it schedulable under a given scheduling policy? FP, EDF, RR ?
- Scheduability analysis is the process of obtaining functional guarantees of a real-time system

PROVIDE GUARANTEE THAT EACH TASK IN THE SYSTEM WILL FINISH EXECUTION WITHIN ITS DEADLINE FOR THE OPERATIONAL LIFE OF THE EMBEDDED SYSTEM !!

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Schedulability: Utilization

Task	C_i	T_i	p_i
τ_1	2	10	0
τ_2	4	15	1
τ_3	5	20	2

- CPU Utilization for task τ_i, $U_i = \frac{C_i}{T_i}$
- $U_1 = \frac{2}{10}$, $U_2 = \frac{4}{15}$, $U_3 = \frac{5}{20}$
- Total Utilization $= U_1 + U_2 + U_3 = 0.72 < 1$
- Since Total Utilization < 1, it is schedulable
- Does not guarantee yet ! Depends on the policy !

Utilization

Is the following task set schedulable by utilization ?

Task	C_i	T_i	p_i
τ_1	3	10	0
τ_2	2	15	1
τ_3	6	15	2
τ_3	2	20	3

Utilization

Is the following task set schedulable by utilization ?

Task	C_i	T_i	p_i
τ_1	3	10	0
τ_2	2	15	1
τ_3	6	15	2
τ_3	2	20	3

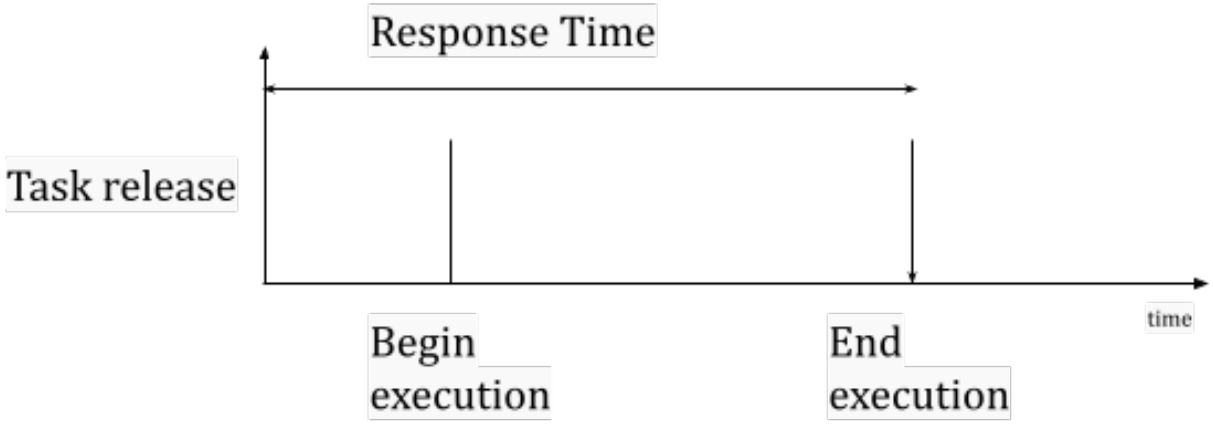
Utilization =

$$\frac{3}{10} + \frac{2}{15} + \frac{6}{15} + \frac{2}{20} = 0.3 + 0.133 + 0.4 + 0.1 = 0.933 < 1$$

Schedulability: Response Time Analysis

- Task arrives at a certain time
- Task begins execution a certain time and then it finishes
- Response Time: it is the time taken for a task to finish execution after its arrival
- For a job J_{ij}, response time R_{ij}:

$$R_{ij} = R_{ij}^{p_{ij}-1} + C_{ij}$$



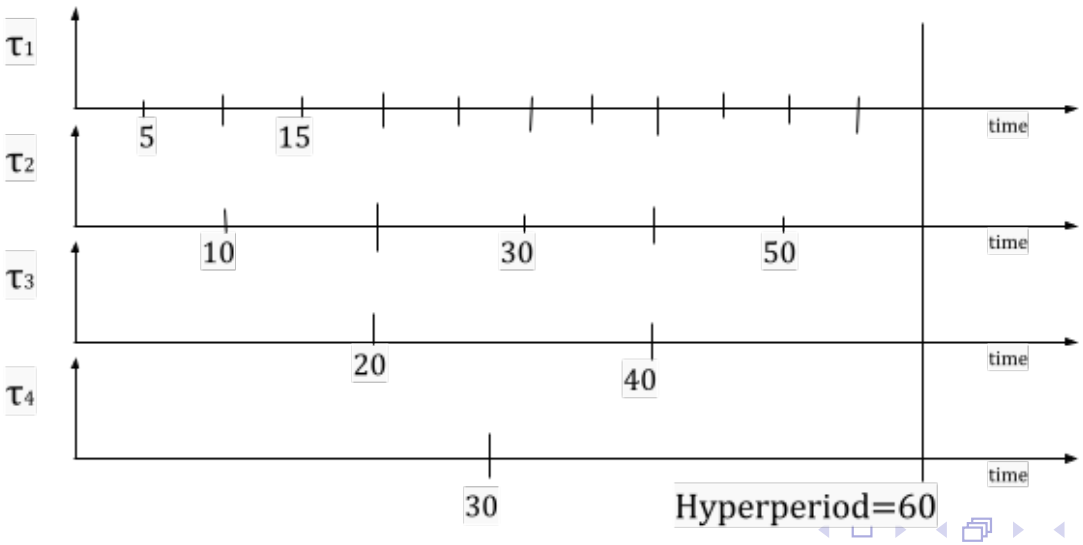
Scheduling decisions: Killing Task

- During execution, if WCET of a task is optimistic, it is possible that the task does not finish by its WCET
- Time allotment for task execution is based in its WCET
- In reality, once task is executing, their is no way of knowing that is has finished until task signals to the scheduler
- If there is no signal and task deadline has arrived, does the scheduler kill the task or allows it to finish ?
- Killing ensures real-time guarantees, not killing ensure functionality !

Schedulability: Hyperperiod

- Hyperperiod= $lcm(T_1, T_2, T_3...)$
- If the
  textbftasks are not allowed to exceed their deadline, Task execution pattern repeats every hyperperiod !

Task	C_i	T_i	p_i
τ_1	2	5	0
τ_2	4	10	1
τ_3	5	20	2
τ_4	5	30	3



Response Time Analysis

Using Response Time Analysis, is the following task set schedulable ?

Task	C_i	T_i	p_i
τ_1	2	5	0
τ_2	4	10	1
τ_3	5	20	2

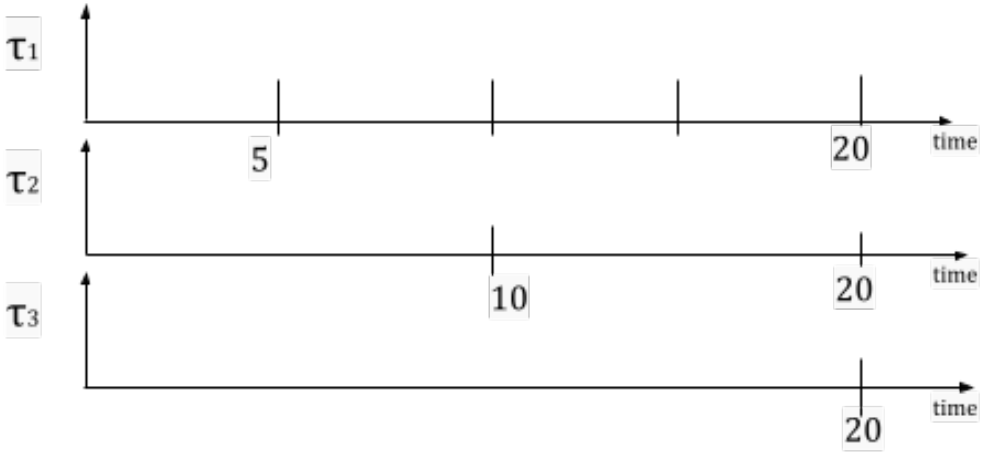
Response Time Analysis

Task	C_i	T_i	p_i
τ_1	2	5	0
τ_2	4	10	1
τ_3	5	20	2

- Hyperperiod ?
- How many jobs does each task has?
- What is the response time for the hyperperiod?

Response Time Analysis

Task	C_i	T_i	p_i
τ_1	2	5	0
τ_2	4	10	1
τ_3	5	20	2

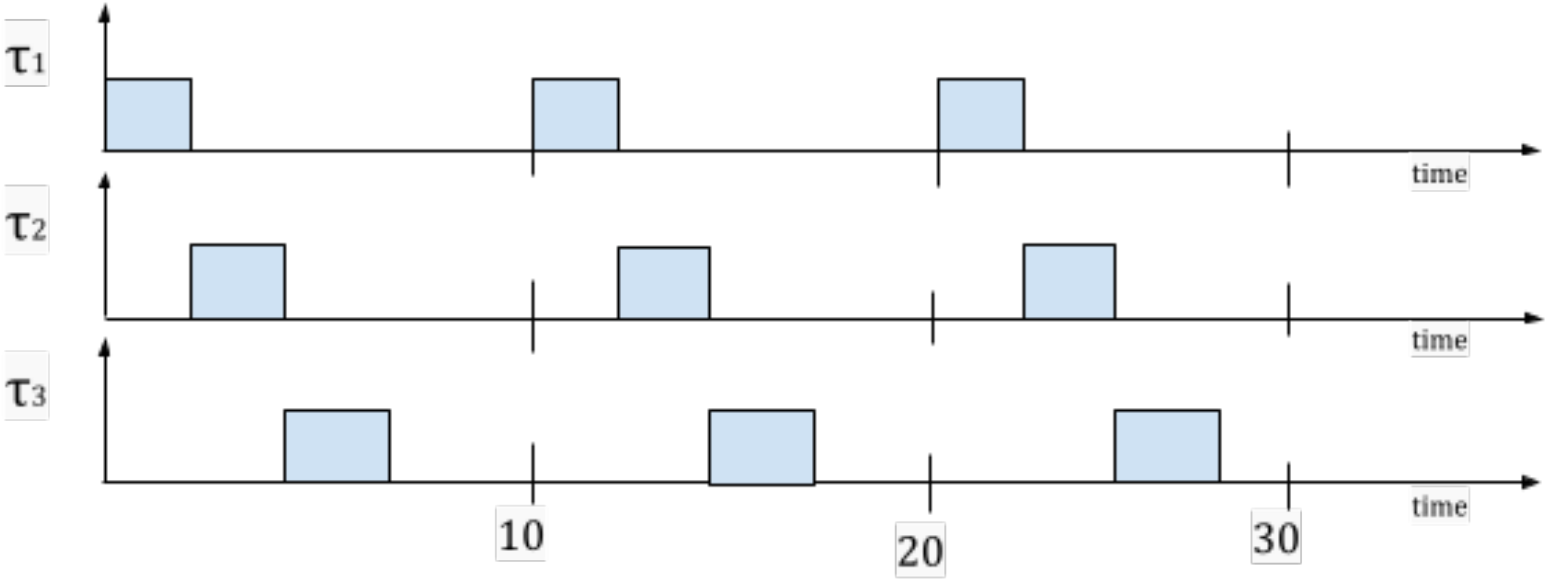


- Hyperperiod =20
- τ_1 has 4, τ_2 has 2, τ_3 has 1
- $RT = 4 * 2 + 2 * 4 + 1 * 5 = 21 > 20$

Round Robin

- Rondo scheme, every task takes turn in fixed order

Task	C_i	T_i	p_i
τ_1	2	10	0
τ_2	3	10	0
τ_3	2	10	0



Round Robin

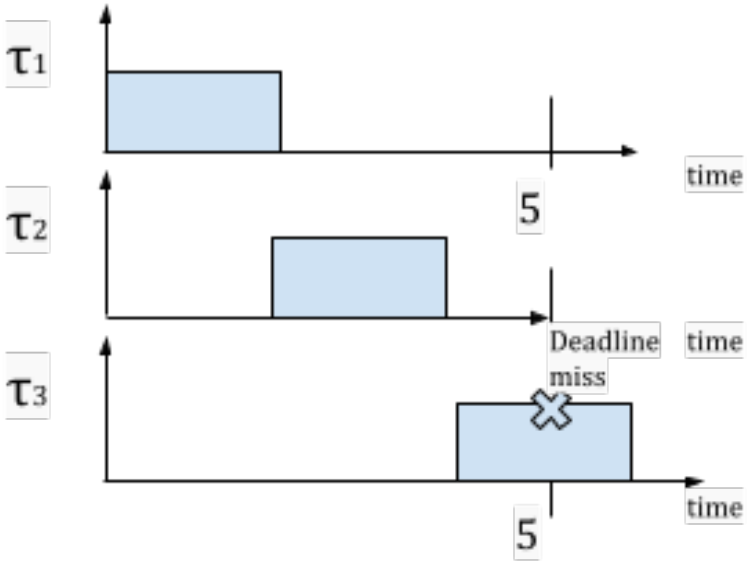
Show if the following task set schedulable using Round Robin?

Task	C_i	T_i
τ_1	2	5
τ_2	2	5
τ_3	2	5

Round Robin

Show if the following task set schedulable using Round Robin?

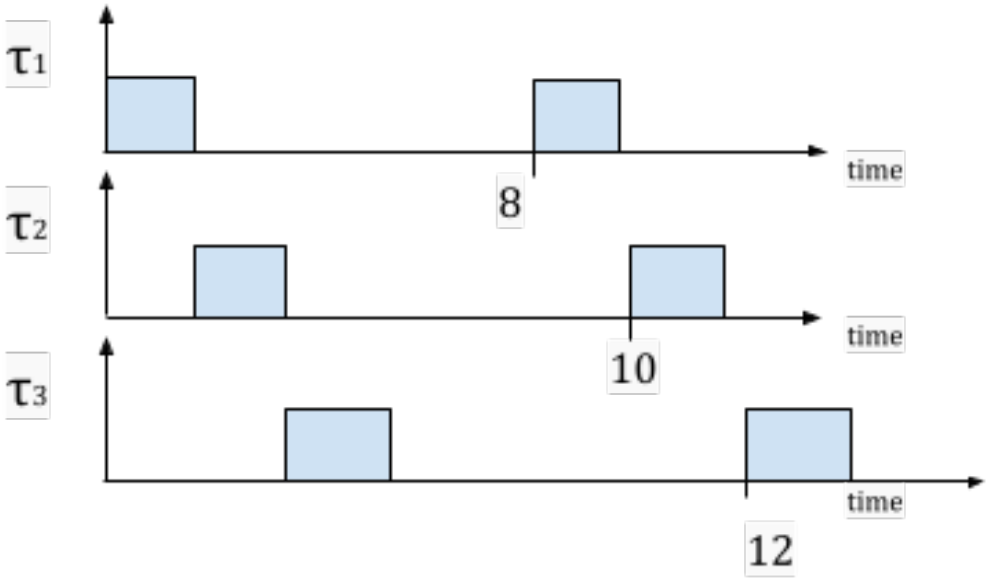
Task	C_i	T_i
τ_1	2	5
τ_2	2	5
τ_3	2	5



Fixed Priority

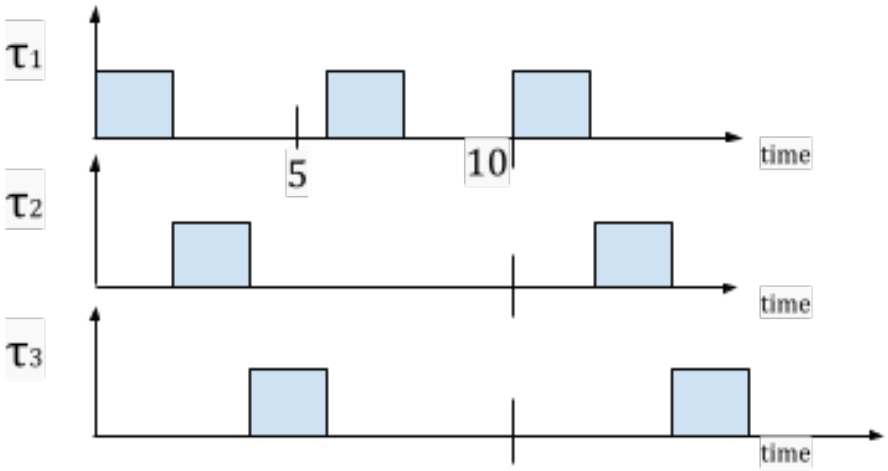
- Each task has a predetermined priority which remains fixed

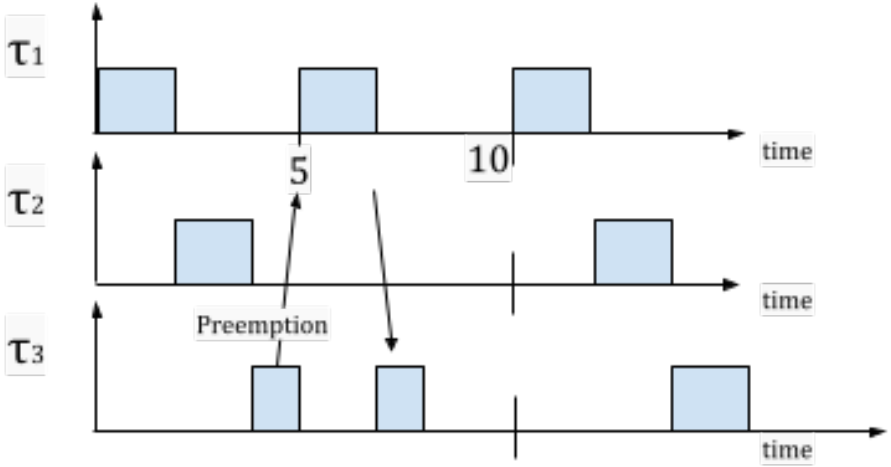
Task	C_i	T_i	p_i
τ_1	2	8	0
τ_2	3	10	1
τ_3	2	12	2



FP Preemptive

Task	C_i	T_i	p_i
τ_1	2	5	0
τ_2	2	10	1
τ_3	2	10	2





Fixed Priority

Show the following task schedulability using FP, Preemptive and Non-Preemptive?

Task	C_i	T_i	p_i
τ_1	2	5	0
τ_2	4	10	1
τ_3	5	15	2

Fixed Priority

Is the following task schedulability using FP, Preemptive and Non-Preemptive? What is the utilization ? What is the hyperperiod ?

Task	C_i	T_i	p_i
τ_1	2	10	0
τ_2	3	15	1
τ_3	2	15	2
τ_3	3	30	3

Fixed Priority

Is the following task schedulability using FP, Preemptive and Non-Preemptive? What is the utilization ? What is the hyperperiod ?

Task	C_i	T_i	p_i
τ_1	3	10	0
τ_2	3	15	1
τ_3	4	15	2
τ_3	7	30	3

Fixed Priority

Is the following task schedulability using FP, Preemptive and Non-Preemptive? What is the utilization ? What is the hyperperiod ?

Task	C_i	T_i	p_i
τ_1	3	10	3
τ_2	3	15	2
τ_3	4	15	0
τ_3	7	30	1

Fixed Priority

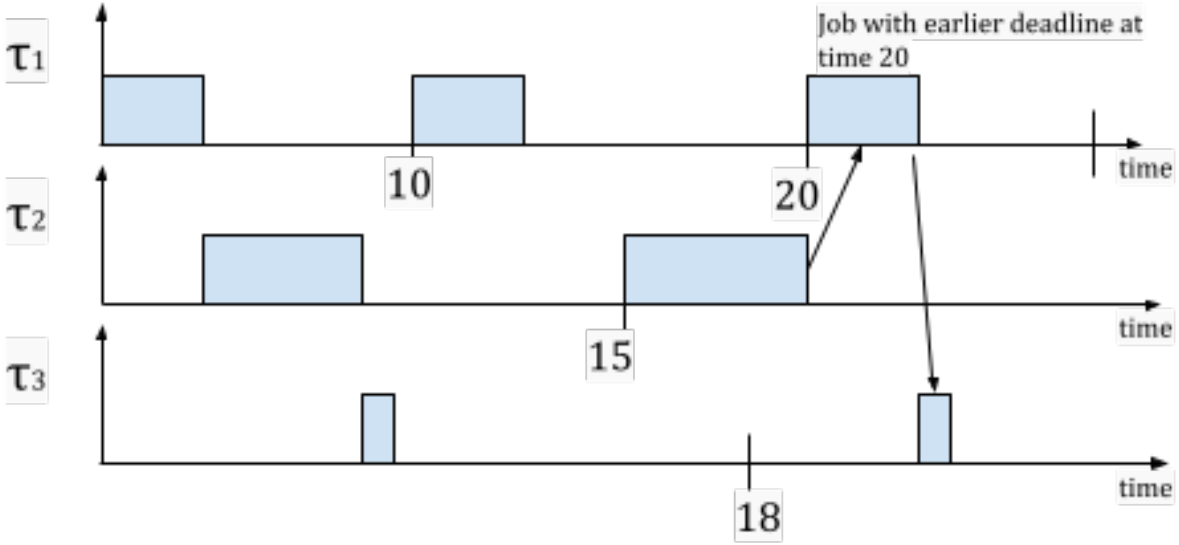
Is the following task schedulability using FP, Preemptive and Non-Preemptive? What is the utilization ? What is the hyperperiod ?

Task	C_i	T_i	p_i
τ_1	2	6	0
τ_2	3	15	1
τ_3	3	15	2
τ_3	5	30	3

Earliest Deadline First

At any given time, an available job with the earliest deadline gets the priority

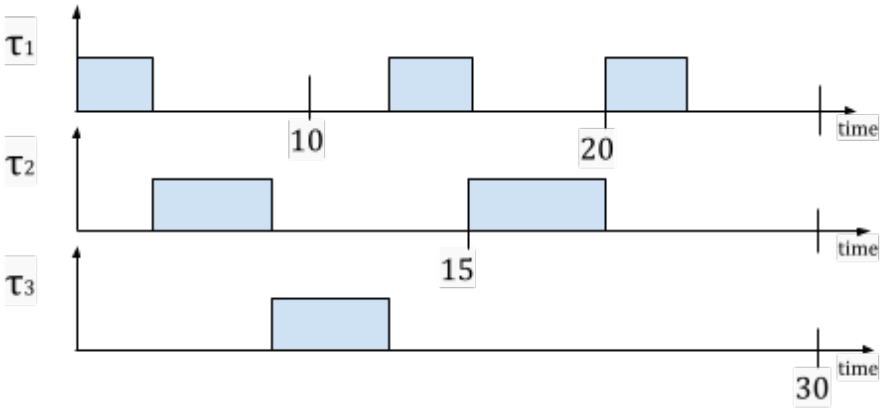
Task	C_i	T_i
τ_1	3	10
τ_2	5	15
τ_3	1	18

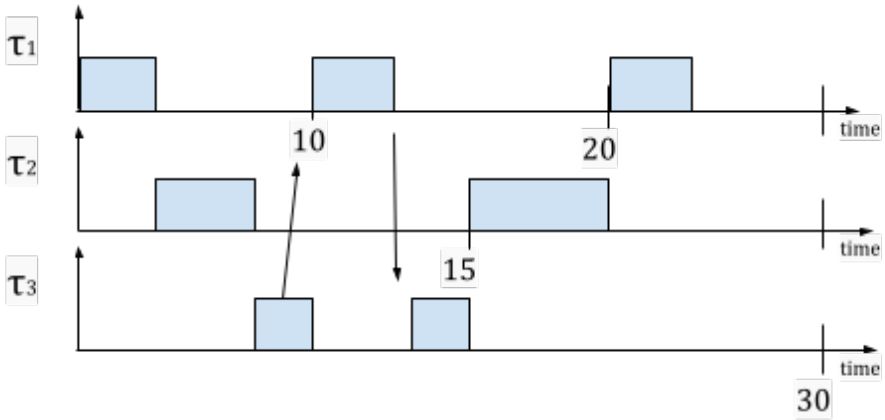


EDF

EDF can be preemptive or non-preemptive

Task	C_i	T_i
τ_1	3	10
τ_2	5	15
τ_3	4	30





EDF

Is the following task set schedulable under EDF scheduling policy ?

Task	C_i	T_i
τ_1	3	10
τ_2	5	15
τ_3	4	30

EDF

Is the following task schedulability using EDF, Preemptive and Non-Preemptive? What is the utilization ? What is the hyperperiod ?

Task	C_i	T_i
τ_1	1	6
τ_2	3	15
τ_3	2	15
τ_3	3	30

EDF

Is the following task schedulability using EDF, Preemptive and Non-Preemptive? What is the utilization ? What is the hyperperiod ?

Task	C_i	T_i
τ_1	3	10
τ_2	3	15
τ_3	3	15
τ_3	4	30

EDF

Is the following task schedulability using EDF, Preemptive and Non-Preemptive? What is the utilization ? What is the hyperperiod ?

Task	C_i	T_i
τ_1	3	10
τ_2	3	15
τ_3	3	15
τ_3	5	30

Utilisation vs Schedule

- Utilisation < 1 is not a sufficient condition for schedulability

Task	C_i	T_i
τ_1	2	5
τ_2	4	7

- Is it schedulable under utilization criteria ?
- What is the hyperperiod ?
- Is the task set is schedulable under
  - Rate Monotonic (shortest deadline has highest priority) ?
  - Earliest Deadline First ?